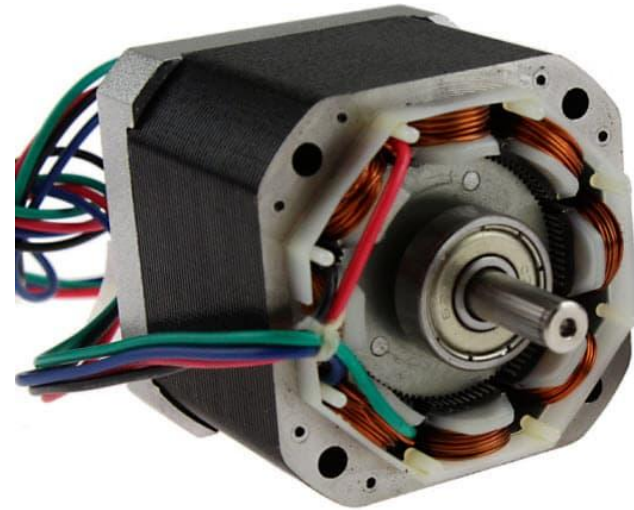
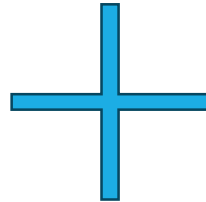
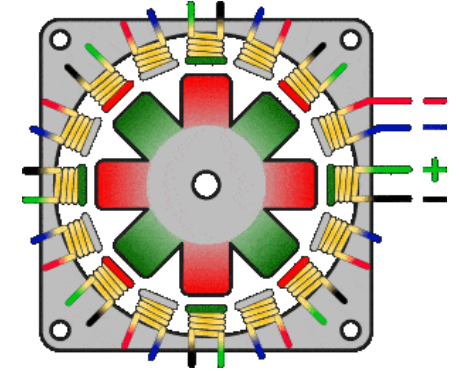
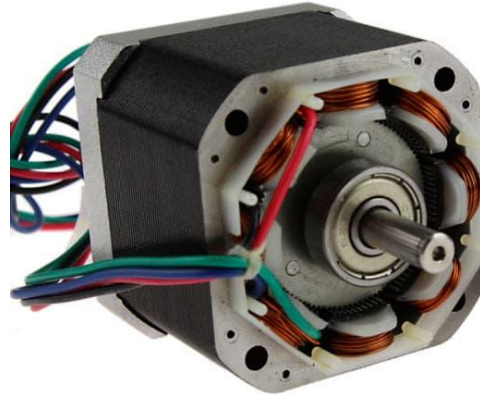
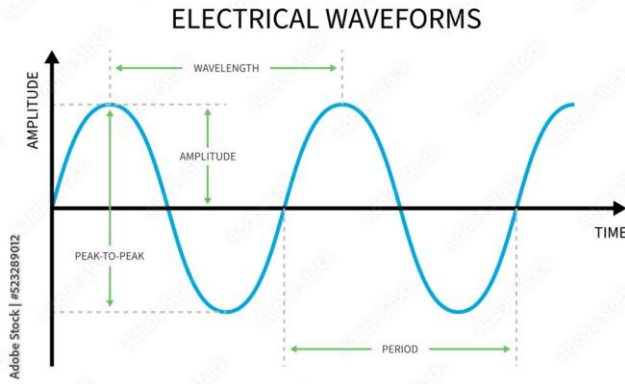


Exp 8: Interfacing an unipolar Stepper motor to LPC2148, to rotate in clockwise/anti-clockwise directions



Intro' to Stepper motor



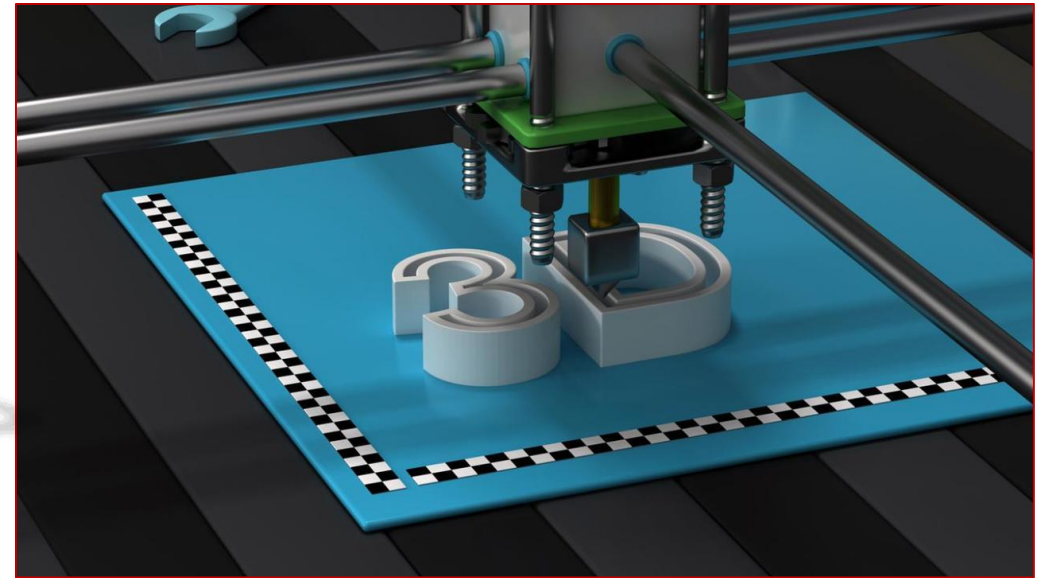
- Stepper motor is an electromechanical device that converts electrical pulses into precise mechanical movements
- These motors are known for their ability to [provide positioning and control](#) without a feedback system
- This makes them ideal for applications requiring precision and reliability
- Applications from [CNC machines](#) to [3D printers](#) where precise control is crucial



Applications of stepper motor



CNC machine – with stepper motor



3D printer – with stepper motor

Unipolar vs. Bipolar stepper motor

Attribute	Unipolar Stepper Motor	Bipolar Stepper Motor
Torque	Lower torque because only half of the winding is used at a time.	Higher torque since the entire winding is utilized during operation.
Precision	Moderate precision due to lower magnetic field utilization.	Higher precision because full coil utilization produces stronger and more stable magnetic fields.
Step Angle	Typically larger step angles (e.g., 7.5° or 15° per step in many models).	Usually smaller step angles (e.g., 1.8° or 0.9°), allowing finer movement.
Accuracy	Lower positioning accuracy due to reduced torque and magnetic control.	Higher positioning accuracy because of stronger torque and better control of rotor position.
Reliability	More reliable in simple systems due to simpler drive circuitry and fewer control requirements.	Slightly more complex control circuitry (H-bridge required), but reliable in high-performance applications.

Unipolar vs. Bipolar stepper motor (contd.)

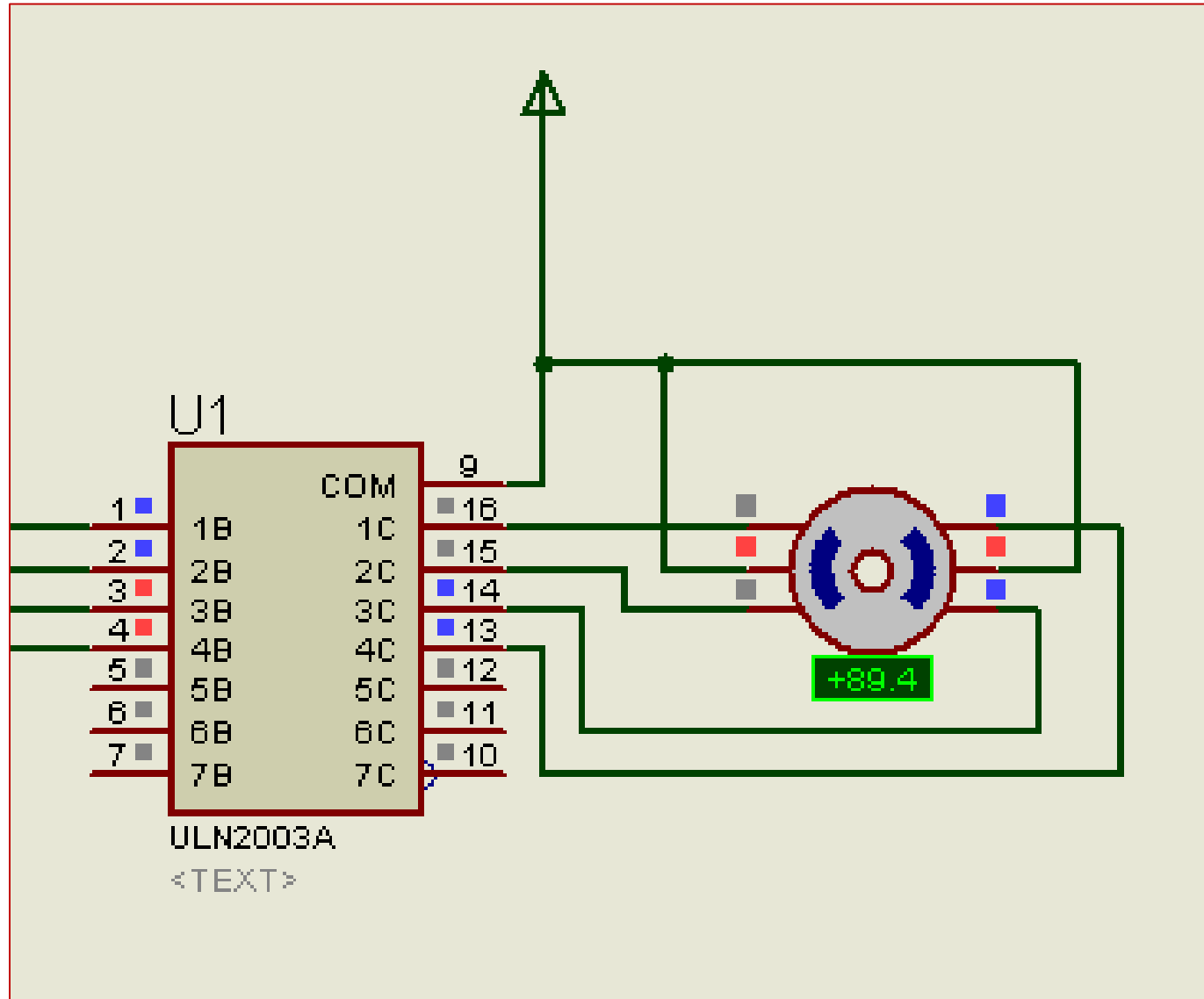
Attribute	Unipolar Stepper Motor	Bipolar Stepper Motor
Number of Wires	Typically 5 or 6 wires due to center taps on each phase winding.	Usually 4 wires, as each phase has a single coil without center taps.
Driver Circuit Requirement	Can be driven using simple transistor arrays such as ULN2003 .	Requires an H-bridge driver (e.g., L293D, L298, A4988) to reverse current direction.
Efficiency	Lower efficiency because only half of the winding is used at any time.	Higher efficiency since the entire winding participates in generating torque.
Control Complexity	Simpler control logic; easier to implement in basic embedded systems labs.	More complex control because current direction must be reversed electronically.
Cost of Driving Circuit	Lower cost due to simpler driver circuitry.	Slightly higher cost because of the need for H-bridge or specialized driver modules.
Typical Applications	Educational kits, low-power positioning systems, simple robotics experiments.	CNC machines, 3D printers, precision robotics, industrial automation systems.

Connecting an unipolar stepper motor to an LPC2148 programmed with Embedded C, using ULN2003A drivers

Component	Proteus Keyword
MCU	LPC2148 (or LPC2138)
Driver	ULN2003A
Stepper Motor	STEPPER MOTOR
Power Supply	VCC
Ground	GND

LPC2148 Pin	ULN2003 Input	Function
P0.0	IN1	Coil A
P0.1	IN2	Coil B
P0.2	IN3	Coil C
P0.3	IN4	Coil D

Connections of components in Proteus



Embedded C program to control an unipolar stepper motor

```
#include <LPC214x.h>

unsigned char step_sequence[4] = {0x03, 0x06, 0x0C, 0x09};

void delay(void)
{
    unsigned int i;

    for(i=0;i<50000;i++); // Adjust for speed
}
```

```
int main()
{ int i;

    IO0DIR |= 0x0F; // P0.0–P0.3 as output

    while(1)
    { for(i=0;i<4;i++)

        { IO0CLR = 0x0F; // Clear previous step

          IO0SET = step_sequence[i]; // Energize coils

          delay(); } } }
```

Energizing the stepper motor coils

